

Online Math Olympiad 2012

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Selected solves from Evan Chen's Online Math Olympiad 2012.

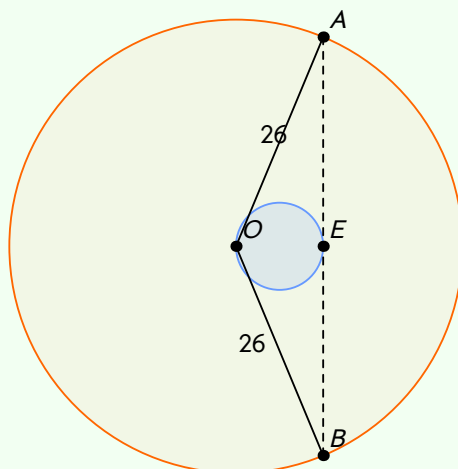
Problem 5

Two circles have radius 5 and 26. The smaller circle passes through center of the larger one. What is the difference between the lengths of the longest and shortest chords of the larger circle that are tangent to the smaller circle?

Solution

The longest chord cannot exceed the diameter of the larger circle which is $2 \times 26 = 52$. This is achievable by taking the diameter of the larger circle which is tangent to the smaller one.

Let O be the centre of the larger circle. Let E be the foot of the perpendicular to AB through O . Denote $h = OE$, $x = BE$ and $y = AE$.



Then

$$26^2 = x^2 + h^2 = y^2 + h^2.$$

So clearly $x = y$ and x is minimised when h is as large as possible. Since $OE \perp AB$, point E is actually the point of tangency from the chord to the smaller circle so OE is a chord of the smaller circle. Thus h cannot exceed 10, giving $x = y = 24$. So the final answer is $52 - (24 + 24) = \boxed{4}$. ■